

Eliot Maes

Data Scientist | AI Engineer | NLP & Applied Machine Learning

+33 6 33 16 21 48 | WP: Europe / Canada | eliotjmaes@gmail.com | LinkedIn: eliot-maes | Github: ejmaes

SUMMARY

ML Engineer / Data Scientist with a PhD in Computer Science and 7+ years of experience in applied machine learning across industry and research. Expertise in NLP, large language models, and multimodal learning. Strong focus on model development, evaluation, and building robust data-driven systems that translate analytical insights into practical solutions.

SKILLS

AI/ML: PyTorch, Keras, Pandas, Polars, Scikit-Learn, HuggingFace Transformers, LangChain

Data Visualization: Tableau Dashboarding, Plotly Dash, Matplotlib, Seaborn, Streamlit

Programming: Python, R, SQL, Vector databases

Web Technologies: HTML/CSS, REST APIs, Web Scraping, Flask

Tools: Linux, Bash/Zsh, Versioning (Git), CI/CD, Docker, LLM prompting

EXPERIENCE

PhD Researcher in NLP

Sept 2021 - Aug 2025

Laboratoire d'Informatique et des Systèmes (LIS)

Marseille, France

- Built predictive models to identify key information exchanges in natural conversations using NLP and multimodal deep learning techniques (Pytorch, HuggingFace, OpenAI), reaching performances similar to human agreement on the corpus studied
- Fine-tuned pre-trained language models (BERT, GPT2) for conversation generation and analysis tasks, optimizing for limited training data scenarios
- Developed zero-shot prediction pipeline using prompt engineering with OpenAI API (GPT-3.5/4) for information exchange detection
- Designed multimodal deep learning architectures combining audio, video, and text data
- Developed the BrainKT corpus: collected, preprocessed and annotated audio, video, and neurophysiological (EEG) data in the multimodal conversational dataset (14 hours) for linguistic and AI research purposes
- 6+ publications in AI/NLP conferences, including LREC-COLING 2024, CoNLL 2022
- Extensive teaching experience in technical subjects including Python programming and data visualization

Data Scientist

Jan 2018 - Aug 2019

Capgemini

Lille, France

- Developed machine learning models for customer churn prediction and built interactive Tableau dashboards for customer behavior analysis for Kiabi, a major French fashion retailer
- Implemented computer vision algorithms (Yolov2) for clothing item recognition in images
- Led internal AI/ML community through project development and community management

IT Consultant Intern

May 2016 - Sept 2016

Wavestone

Paris, France

- Wavestone IT System Department, ERP Upgrade Project, QA engineering team

EDUCATION

PhD in Computer Science

2021 - 2025

LIS, Aix-Marseille Université

Marseille, France

Research topic: Multimodal Learning for information transfer modeling in natural conversation

MSc in Cognitive Science

2019 - 2021

Aix-Marseille Université

Marseille, France

Curriculum focus: Language, Brain and Cognition

Research Internships: Developed predictive models for conversation analysis in parent-child and human-robot interactions

RECENT PROJECTS

Japanese Learning iOS App

SwiftUI, MLX

- JLPT study app covering kanji, vocabulary and grammar, with practice tests, parsed news & books for in-context vocab lookup
- On-device chat with user-selectable MLX models
- Merged and normalized JLPT content from multiple sources into a unified dataset shipped in-app; optimized data structures for efficient on-device querying

Research Tab Manager

Docker, FastAPI, PineconeDB

- Ingests Safari tabs, extracts article content and runs local LLM summarisation
- Streaming RAG/QA over the personal knowledge base with cross-encoder reranking (Hybrid BM25 and semantic search with Reciprocal Rank Fusion)

LANGUAGES

French: Native proficiency

English: Fluent

Japanese: Conversational

German: Conversational

PUBLICATIONS

Maës, Eliot, Leonor Becerra-Bonache, and Philippe Blache. "Locating strongly informative utterances in conversation using multimodal cues". In: *CogSci 2025*. 2025.

Maës, Eliot, Hossam Boudraa, Philippe Blache, and Leonor Becerra-Bonache. "Did You Get It? A Zero-Shot Approach To Locate Information Transfers In Conversations". In: *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. 2024.

Full list available at ejmaes.github.io